

Appendix G

Committer for 1d diffusive barrier crossing

In section 15.4 we discussed the concept of the committer that quantifies the fraction of trajectories in a certain part of configuration space that will go on to reach the product state B .

Here, we give a very simple example of such a calculation for the case of a diffusive crossing of a one-dimensional barrier of width L . Mathematicians will cringe at the derivation below because it is clumsy, and the final result can be obtained in one line (literally¹) using the backward Kolmogorov equation in the theory of stochastic differential equations [637].

G.1 1d diffusive barrier crossing

Consider a process whereby a system diffusively crosses a one-dimensional barrier of constant height $\epsilon \ll k_B T$ and width L . The diffusion constant of the system is assumed to be constant and equal to D . We denote the coordinate of the system by x . The barrier is in the x -range $0 \leq x < L$. We call region $x < 0$ the “reactant” region A . The product region (B) corresponds to $x > L$.

What is the probability that a system at point x_0 in the barrier region will “commit” to state B ? To answer this question, we must solve the 1d diffusion equation in the barrier region and compute the integrated flux to B . If $\epsilon/k_B T$ is large enough, all systems arriving at $x = L$ will proceed to B , and all systems arriving at $x = 0$ will proceed to A . We can therefore treat both $x = 0$ and $x = L$ as absorbing boundaries, meaning that the concentration $c(x)$ always vanishes at these boundaries.

The diffusion problem is best solved by Fourier transforming. The diffusion equation that we must solve is of the form:

$$\dot{c}(x, t) = D \left(\frac{\partial^2 c(x, t)}{\partial x^2} \right). \quad (\text{G.1.1})$$

¹ The one line is $\left(\frac{\partial^2 C(x)}{\partial x^2} \right) = 0$, where the committer $C(x)$ satisfies the boundary conditions $C(0) = 0$ and $C(L) = 1$. But to get there would have required a mini-course on stochastic differential equations, an investment well worth the effort but beyond the scope of this book.

We expand $c(x, t)$ in the region $0 < x \leq L$ in Fourier modes. As the concentration must vanish at the boundaries, we only consider sine components²:

$$c(x, t) = \sum_{n=1}^{\infty} a_n(t) \sin(n\pi x/L), \quad (\text{G.1.2})$$

and hence

$$a_n(t) = \frac{2}{L} \int_0^L dx \, c(x, t) \sin(n\pi x/L). \quad (\text{G.1.3})$$

In what follows, we will use the notation $q_n \equiv n\pi/L$. We assume that the concentration profile at $t = 0$ is a delta-function at x_0 :

$$c(x, 0) = \delta(x - x_0), \quad (\text{G.1.4})$$

then

$$a_n(0) = \frac{2}{L} \sin(q_n x_0). \quad (\text{G.1.5})$$

In Fourier components, the diffusion equation reads:

$$\dot{a}_n(t) = -Dq_n^2 a_n(t) \quad (\text{G.1.6})$$

and hence:

$$a_n(t) = a_n(0) \exp(-Dq_n^2 t). \quad (\text{G.1.7})$$

G.2 Computing the committor

The probability current at $x = L$ is

$$j(L, t) = -D \left(\frac{\partial c(x, t)}{\partial x} \right)_L = -D \sum_{n=1}^{\infty} a_n(t) q_n \cos(q_n L). \quad (\text{G.2.1})$$

The integrated flux to B is the probability that a system originally at x_0 will reach the absorbing boundary at L . In other words, it is the committor $C(x_0)$, given by:

$$C(x_0) \equiv \int_0^{\infty} dt \, j(L, t) = -\frac{2}{L} \sum_{n=1}^{\infty} \sin(q_n x_0) \frac{\cos(q_n L)}{q_n}. \quad (\text{G.2.2})$$

Rather than solving this equation, we consider the derivative

$$\left(\frac{\partial C(x_0)}{\partial x_0} \right) = -\frac{2}{L} \sum_{n=1}^{\infty} \cos(q_n x_0) \cos(q_n L), \quad (\text{G.2.3})$$

² The sine expansion applies to a periodic, anti-symmetric ($f(-x) = -f(x)$) function with period $2L$. We are only interested in the behavior for $0 \leq x < L$.

but the right-hand side is nothing else than the Fourier transform of $-\delta(L - x_0)$, excluding the $n = 0$ part.³ Therefore, we can write $\left(\frac{\partial C(x_0)}{\partial x_0}\right)$ as

$$\left(\frac{\partial C(x_0)}{\partial x_0}\right) = \frac{1}{L} - \delta(L - x_0). \quad (\text{G.2.4})$$

As $\delta(L - x_0) = 0$ for $x_0 \neq L$, the value of $\left(\frac{\partial C(x_0)}{\partial x_0}\right)$ equals $1/L$ for $x_0 < L$. Integrating the above equation with respect to x_0 , and using the fact that $C(0) = 0$, we get

$$C(x_0) = \frac{x_0}{L}. \quad (\text{G.2.5})$$

In other words, the value of the committer for this simple barrier crossing equals x_0/L . It varies from 0 at $x_0 = 0$ to 1 at $x_0 = L$. This is just an example that shows how committers may be computed if we know the solution of the Fokker-Planck equation of the problem.

³ The cosine expansion applies to a periodic, symmetric ($f(-x) = f(x)$) function with period $2L$. Again, we are only interested in the behavior for $0 \leq x < L$.